

quantum safe: writeup (lattice ver)

we're given a bunch of ciphertexts defined as

$$\vec{c}_i = \vec{m}_i \cdot A + \vec{r}$$

where $\vec{m} = (x, y, z)$ where:

- $x = \text{ord}(\text{flag}_i)$
- $y, z \in \{0..100\}$

and $\vec{r}$ is all numbers $\in 0..10$. it's really important to note that $\vec{r}$ is fixed for all ciphertexts from here we can see that

$$\vec{c} = (x, y, z) \cdot A + \vec{r}$$

$$\vec{c} = x\vec{A}_1 + y\vec{A}_2 + z\vec{A}_3 + \vec{r}$$

we could aim to find all (x, y, z) independently, or better yet, find $\vec{r}$. if we find $\vec{r}$, we can do

$$(\vec{c}_i - \vec{r}) \cdot A^{-1} = \vec{m}_i$$

now, we know that the flag must start with HTB{. so, in our case,

$$\vec{c} - x\vec{A}_1 = y\vec{A}_2 + z\vec{A}_3 + \vec{r}$$

we can dissect this equation as:

1. $\vec{c} - x\vec{A}_1$: known
2. $y\vec{A}_2 + z\vec{A}_3$: vectors known, coefs unknown
3. $\vec{r}$: unknown

it's interesting because since we know the vectors in (2), we can make a lattice $\mathcal{L} = \{\vec{A}_2, \vec{A}_3\}$, and see that $y\vec{A}_2 + z\vec{A}_3$ is in that lattice

so, **since $\vec{r}$ is small** we could solve CVP to find the closest vector to $\vec{c} - x\vec{A}_1$, which should be $y\vec{A}_2 + z\vec{A}_3$. so we just do:

$$(\vec{c} - x\vec{A}_1) - (y\vec{A}_2 + z\vec{A}_3) = \vec{r}$$

conclusions

- I misnomered $\vec{m} \cdot A$ with $A \cdot \vec{m}$, thinking $\vec{m}$ and $\vec{r}$ could be thought as vertical vectors, which tripped me up a lot. I should be more careful about left vs right matrix multiplication
- This exercise shows a very nice way to see what is a CVP versus an SVP one!